SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

Total Marks: 50

Annexure A

Code of Conduct:

I Karnam Bhaskar Gnanesh(192324080) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
K B Gnanesh

Annexure A

1. Secure Password Storage in Web Applications
A web application stores user passwords and needs to protect them against database breaches.

A. Explain why storing plaintext passwords is insecure.
B. Suggest a secure method for storing passwords and justify your choice.

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2. Digital Signature for Document Authentication
A company wants to ensure that official documents sent electronically are authentic and unaltered.

A. Explain how digital signatures can be used in this scenario.
B. Justify how this method ensures both integrity and non-repudiation.

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3. Preventing Replay Attacks in Online Transactions
An online payment system is vulnerable to replay attacks where attackers resend previously captured transactions.

A. Explain how replay attacks occur.
B. Suggest a cryptographic technique to prevent such attacks and justify your answer.

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4. Secure Communication over Public Wi-Fi
A user accesses sensitive information using a public Wi-Fi network without encryption.

A. Identify the risks involved in this scenario.
B. Recommend a secure communication method and justify how it protects user data.

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5. Data Integrity in Cloud Storage
An organization stores critical data in a cloud server and wants to ensure that the data is not altered.

A. Explain how hashing can be used to verify data integrity.
B. Suggest an additional mechanism to enhance trust and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Secure Password Storage in Web Applications
A. Explain why storing plaintext passwords is insecure.

Storing plaintext passwords is insecure because anyone who gains unauthorized access to the database can read users' passwords directly. Attackers may use these passwords to access other accounts if users reuse the same password. This can lead to identity theft, financial loss, and privacy violations.

B. Suggest a secure method for storing passwords and justify your choice.

A secure method is to use Argon2id password hashing with a unique salt for each user. Argon2id converts the password into a one-way hash, making it difficult for attackers to recover the original password. The salt prevents attackers from identifying users with identical passwords and makes precomputed rainbow-table attacks ineffective. During login, the system hashes the entered password using the stored parameters and salt, then verifies the result.

2. Digital Signature for Document Authentication
A. Explain how digital signatures can be used in this scenario.

A digital signature can verify the authenticity of official electronic documents. The sender generates a hash of the document and signs that hash using their private key. The receiver uses the sender's public key to verify the signature and compares the verified document hash with a newly calculated hash of the received document.

B. Justify how this method ensures both integrity and non-repudiation.
- Integrity: If the document is modified after signing, its hash changes, and signature verification or hash comparison fails.
- Non-repudiation: A valid signature provides evidence that the corresponding private key was used to sign the document. When the private key is securely controlled and the signer's identity is reliably verified, it helps prevent the signer from denying the signature.

3. Preventing Replay Attacks in Online Transactions
A. Explain how replay attacks occur.

A replay attack occurs when an attacker captures a legitimate transaction request and sends the same request again without authorization. For example, an attacker may resend a captured payment request to attempt to charge a customer's account more than once.

B. Suggest a cryptographic technique to prevent such attacks and justify your answer.

A suitable technique is to use message authentication codes (HMACs) with a unique nonce and timestamp. The sender includes the transaction details, nonce, and

timestamp in the authenticated message. The receiver verifies the HMAC, checks that the timestamp is within an acceptable period, and rejects previously used nonces.

This prevents attackers from successfully reusing captured requests. The payment system should also use unique transaction IDs and idempotency checks to ensure that the same payment is not processed more than once.

4. Secure Communication over Public Wi-Fi

A. Identify the risks involved in this scenario.

Using public Wi-Fi without encryption exposes users to several security risks:

1. Eavesdropping: Attackers may intercept sensitive information transmitted over the network.
2. Man-in-the-middle attacks: An attacker may intercept and potentially modify communications between the user and a website.
3. Credential theft: Login credentials and personal information may be exposed.
4. Session hijacking: Attackers may steal session tokens and gain unauthorized access to accounts.
5. Fake Wi-Fi hotspots: Fraudulent networks may imitate legitimate public Wi-Fi to collect user information.

B. Recommend a secure communication method and justify how it protects user data.

The recommended method is HTTPS using TLS (Transport Layer Security). HTTPS encrypts data transmitted between the user's browser and the website, protecting it from eavesdropping and unauthorized modification. TLS also authenticates the website through digital certificates, helping users avoid connecting to impersonated websites.

Users should verify the website address and certificate, avoid ignoring browser security warnings, and use a trusted VPN when appropriate. A VPN provides an additional encrypted tunnel between the device and the VPN server, but it does not replace HTTPS.

5. Data Integrity in Cloud Storage

A. Explain how hashing can be used to verify data integrity.

Hashing can verify whether stored data has been altered. The organization calculates a cryptographic hash, such as SHA-256, for the original file and securely stores the hash value. Later, it downloads the file and calculates its hash again. If both hash values match, the file is consistent with the original data with very high confidence. If they differ, the file may have been modified or corrupted.

B. Suggest an additional mechanism to enhance trust and justify your answer.

An additional mechanism is to use a digital signature to authenticate the stored hash. The organization signs the hash using its private key and stores the signature separately or in a protected location. During verification, the public key is used to check the signature and confirm that the hash was signed by the authorized organization.

This provides both integrity verification and authenticity. Even if an attacker modifies the cloud file and replaces its stored hash, the attacker cannot create a valid signature without the private key. The private key must therefore be securely protected.

Result: The five scenarios demonstrate how password hashing, digital signatures, replay protection, TLS encryption, and cryptographic integrity verification help protect information and secure digital communications.